

PREDICTIVE CAPTURE: A SINGLE STATISTIC PRICES THE DECODABILITY—USABILITY GAP ACROSS RESERVOIR FAMILIES

A P R E P R I N T

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ABSTRACT

Three earlier findings in this research program each uncovered a version of the same gap: a recurrent reservoir can *store* information, recoverable by a linear probe, without being able to *use* it in a trained readout — an addressed-memory echo state network (ESN) that decodes deeper into its input history yet predicts next characters worse, a bridged ESN whose future-keyed representational geometry adds nothing to its trained readout's accuracy, and a state-quantization dial that degrades bits per character (bpc) smoothly while decode structure persists. This study asks whether that gap is one measurable quantity or three unrelated coincidences. We define **predictive capture** $x_c = 1 - R^2_p$: the fraction of a fixed 5-gram language model's next-character distribution that a linear map of a reservoir state fails to reconstruct, measured out-of-sample. The two-parameter law $bpc = \text{intercept} + \beta \cdot x_c$, calibrated on exactly two cells of a state-quantized ESN, was applied blind to 15 held-out cells spanning three reservoir families that had never before shared a common evaluation. On two independently drawn random seeds: every risk-bearing sign prediction correct (4/4 both seeds), Spearman(predicted, measured) 0.991 and 0.980, median absolute error 0.043 and 0.041 bits, and a purpose-built analytic null beaten by an order of magnitude on both metrics. Stronger: seed 0's already-fitted constants, applied to seed 1's data with zero refitting, clear the same thresholds (Spearman 0.980, median error 0.0445) — the fitted β is value-stable across independently drawn reservoirs (2.8076 vs 2.8174, a 0.35% difference). Two seeds is treated as a floor, not a sufficient result: the program's own independent audit of this entry is still pending, and a third seed is the named next step toward greater confidence.

Keywords reservoir computing · echo state networks · linear probes · decodability vs. usability · predictive capture · cross-family generalization · pre-registration

§1 The question

A recurring result in this program is a dissociation: instruments that measure what a reservoir state *stores* (linear decode depth, geometric alignment with a target) and instruments that measure what a trained readout can *use* (validation bits per character) rank the same reservoirs differently, in reproducible ways. Three separate lines of experiments encountered this independently, each through

its own mechanism — content-addressed memory, a future-keyed linear bridge, and state quantization. Either the program had found three different phenomena, or one phenomenon three times. The distinction is testable: if it is one phenomenon, there should exist a single statistic, computable from a reservoir's states without training any readout, that prices all three families' bpc on a single line.

§2 Method

No new reservoir dynamics are introduced. This study reuses the construction code verbatim from the three source experiment families and re-evaluates 17 cells on identical text8 splits [3] (2M training characters, $N=1024$ -unit reservoirs [1]): six cells from the addressed-memory ESN family ({Hadamard-permutation addressing, sparse random mixing} $\times$ spectral radius $\rho \in \{0.6, 0.95, 1.1\}$), four low- ρ crossover arms ({permutation addressing, sparse random mixing} $\times$ $\rho \in \{0.45, 0.8\}$), four cells from the state-quantized ESN family ($b \in \{1, 2, 4, \text{float32}\}$ bits per unit), and three arms of the future-keyed linear bridge (forward state x , bridge-mapped state $F(x)$, and their concatenation $[x \parallel F(x)]$). An in-run consistency gate required every recomputed ridge-regression validation bpc to match its archived value to three decimal places — 0 failures in 17 cells at seed 0; at seed 1, all 7 cells with a matching same-seed archive matched exactly, and the remaining 10 had no same-seed archive to gate against, exactly as disclosed before running.

```
# STORAGE INSTRUMENT: predictive capture, no readout training required
#   P5 = fixed 5-gram language model (additive-smoothed, 5M training chars)
#   fit ridge regression: state(t) -> log2 P5[context_4(t)] (27-dim target)
#   R2_p = pooled out-of-sample R2 of that fit
x_c = 1 - R2_p

# THE LAW: two parameters, calibrated on two cells
#   (state-quantized reservoir at b=1 and float32 only)
bpc_predicted = intercept + beta * x_c
# then applied blind to the 15 held-out cells
```

The pre-registered control is an analytic null with real statistical teeth: a credit-apportioned n -gram value, which credits each character decodable at lags 1–4 with the marginal n -gram value of each context extension, telescoping to the 5-gram anchor under perfect decoding.

§3 Pre-registered predictions

- **O1** (ordinal): the sign of Δx_c matches the sign of Δbpc in the four risk-bearing matched pairs (addressed-memory permutation addressing vs. sparse mixing at $\rho \in \{0.6, 0.95, 1.1\}$; quantized $b=2$ vs. float32). The two bridge-arm pairs are nested-model consistency checks, registered as such and excluded from scoring. Loss condition: ≥ 2 pairs fail.
- **O2** (ordinal): Spearman(predicted, measured) ≥ 0.8 over the 15 held-out cells; a pre-registered gray zone of $[0.6, 0.8)$ would promote nothing either way; < 0.6 rejects the single-law reading.
- **Q1** (quantitative): median $|\text{error}| \in [0.00, 0.08]$ bits; > 0.15 rejects the quantitative law even if the ordinal predictions survive.

- **Named overall loss:** the analytic null matches or beats the law on O1 and O2.

The seed-1 replication (predictions E1–E4) re-registered the same thresholds for a freshly fitted law, and added a sharper test, E2: seed 0's already-fitted parameters applied to seed 1's cells with *zero refitting* must clear the same thresholds. One scope reduction was disclosed before running: O1 is scored on raw sign-match of point estimates rather than the originally specified bootstrap interval on Δx_c — an O1 pass is accordingly somewhat softer evidence than the original registration envisioned, and is labeled as such throughout.

§4 Results

Table 1. Both random seeds (internal run log #29 / #30). All pre-registered thresholds cleared; no gray zone approached. The null's Spearman correlation is seed-invariant by construction.

metric	seed 0	seed 1	registered threshold
O1 sign matches (risk-bearing pairs)	4/4	4/4	≥2 fail = loss
Spearman(predicted, measured), 15 cells	0.991	0.9801	≥ 0.8
median bpc_predicted - bpc_measured (bits)	0.0427	0.0409	≤ 0.08
fitted β (intercept 2.0614, both seeds)	2.8076	2.8174	–
frozen seed-0 law applied to seed-1 data (E2)	–	0.9801 / 0.0445	same thresholds
analytic null: Spearman / median error	0.3745 / 0.9483	0.3745 / 0.9437	must lose

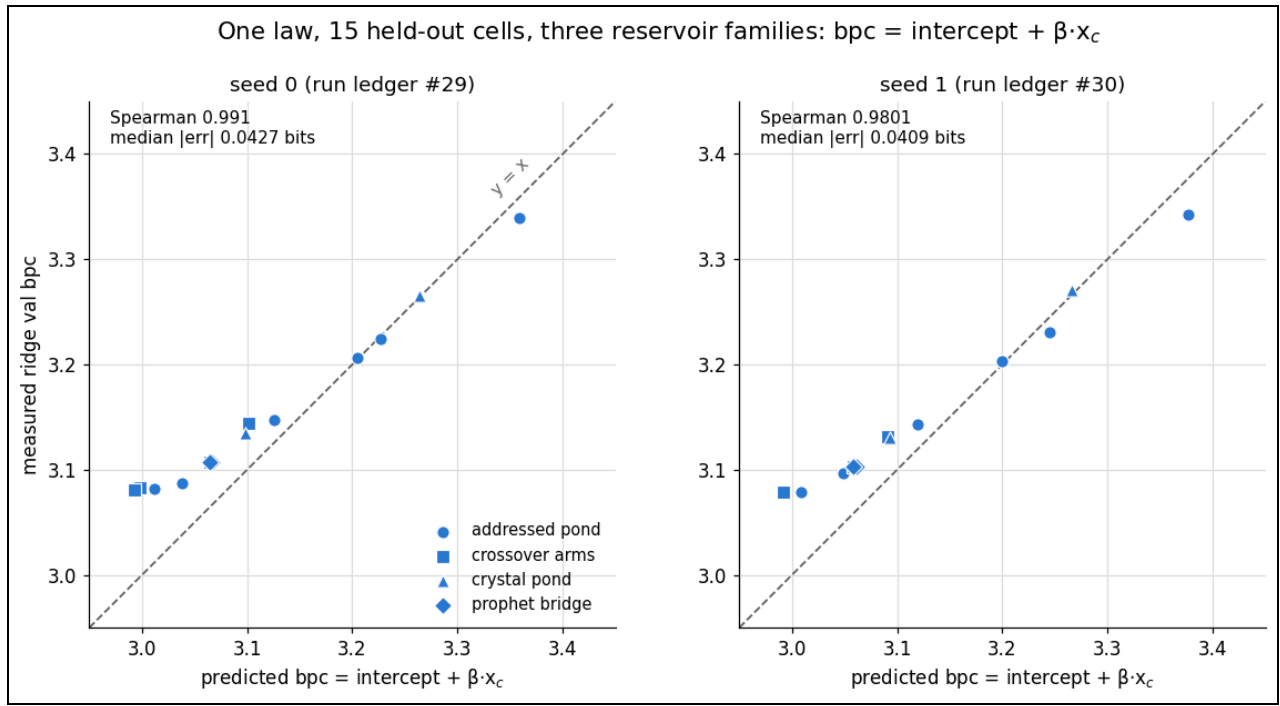


Figure 1. Predicted versus measured ridge-regression validation bits per character, both seeds (internal run log #29, #30). Each panel: 15 held-out cells from three reservoir families; the two calibration cells (state-quantized ESN at $b=1$ and float32) are not shown. The dashed line is $y = x$, not a fit.

Every pre-registered threshold cleared, on both seeds, with no gray zone approached.

O1: 4/4 sign matches both seeds — for example, at seed 1 the addressed-memory $\rho=1.1$ pair has $\Delta x_c = +0.0632$ against $\Delta \text{bpc} = +0.1401$, and the quantized $b=2/\text{float32}$ pair has $\Delta x_c = +0.0736$ against $\Delta \text{bpc} = +0.1678$. O2 and Q1: Spearman 0.991 / 0.9801, median error 0.0427 / 0.0409 bits. The analytic null collapsed to the 5-gram anchor almost everywhere — per-lag decode accuracy saturates near 1.0 at lags 1–4 for these reservoir families, exactly the pre-registered failure mode — and lost by roughly an order of magnitude on both metrics, both seeds.

E2, the stronger result: the same fitted constants generalize to an independently drawn reservoir. Seed 0's frozen constants, applied blind to seed 1's fifteen cells, score Spearman 0.9801 and median error 0.0445 bits — a cost of only 0.0036 bits for using the "wrong" seed's parameters. The law is value-stable, not merely stable in functional form.

Two secondary findings travel with this result, both confirmed at both seeds. First, the 5-gram model's own in-run validation bpc is 2.0614 — bit-identical across seeds, and a stable ~ 0.098 bits below the "2.159" value documented in the program's standing configuration; the law was fit and scored against the in-run value throughout, and the documented value should not be quoted without this caveat until reconciled. Second, a previously established low- ρ crossover in this program (permutation addressing outperforming sparse random mixing below $\rho^* \approx 0.72$) is specific to a *logistic-regression* readout: under ridge regression — this study's usability instrument — permutation addressing underperforms sparse mixing at every tested ρ from 0.45 to 1.1, both seeds, with no crossover, and x_c tracks ridge's monotonic ordering throughout. One caveat on that second finding: at seed 1 the $\rho=0.45$ ridge gap narrowed to 0.0008 bits — the sign did not flip, but that specific sub-claim now rests on a near-tie and would need a third seed to firm up.

§5 Discussion

Three gaps were one gap. The dissociations this program kept encountering independently — storage without usability, representational geometry without payoff, quantization as a pure bpc dial — are priced by a single scalar computed from reservoir states alone, using the same two fitted constants, across sparse-mixing reservoirs, content-addressed memories, quantized reservoirs, and a future-keyed linear bridge. Within this inventory, "how much a linear readout of the state misses of what a 5-gram language model already knows" is the decodability–usability gap, to a median accuracy of roughly one-twentieth of a bit.

What this does not yet establish. Uncertainty is reported as a min–max range across two seeds rather than a bootstrap interval; this study's own replication count is 2, though it re-scores cells drawn from source experiments with substantially deeper replication histories. This research program requires an independent audit of every promoted finding — re-deriving these numbers from the raw result files using independently written code — and that audit has not yet been run for this result; the finding is reported here as replicated and audit-pending, per the program's own rules. A pre-registered blind extension of the frozen law to further, previously unseen reservoir families is under study; no claim from that extension is made here. A third seed would move the law from "replicated" to "difficult to doubt," and is the named next step.

§6 References

1. H. Jaeger, "The 'echo state' approach to analysing and training recurrent neural networks," GMD Report 148, German National Research Center for Information Technology, 2001.
 2. M. Lukoševičius, H. Jaeger, "Reservoir computing approaches to recurrent neural network training," *Computer Science Review* 3(3), 2009.
 3. M. Mahoney, "Large text compression benchmark" (text8), matmahoney.net/dc/textdata.
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§7 Provenance

- **f2b91d28** — 2026-08-11-assay-office: 17 cells + bridge arms, seed 0 (internal run log #29). Pre-registered before running; O1, O2 and Q1 all confirmed; internally rated at the program's highest first-pass significance tier, which by standing policy forbids publication until a replication passes — the seed-1 rerun was queued ahead of all other pending work.
- **86b4occ1** — 2026-08-11-assay-office-rerun1: seed 1 (internal run log #30). E1 (fresh fit) fully confirmed; E2 (frozen seed-0 law, zero refitting) confirmed; E3 (intercept bit-identical) confirmed; E4 (ridge/logistic crossover split) confirmed, with the $\rho=0.45$ near-tie flagged. Promoted to the program's confirmed-findings record per the pre-stated promotion rule.

All claims are judged strictly against the pre-registered wording; disclosed scope reductions are noted above. 2 seeds; replicated before publication; the program's independent audit of this entry is still pending. Internal designation: assay office.

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